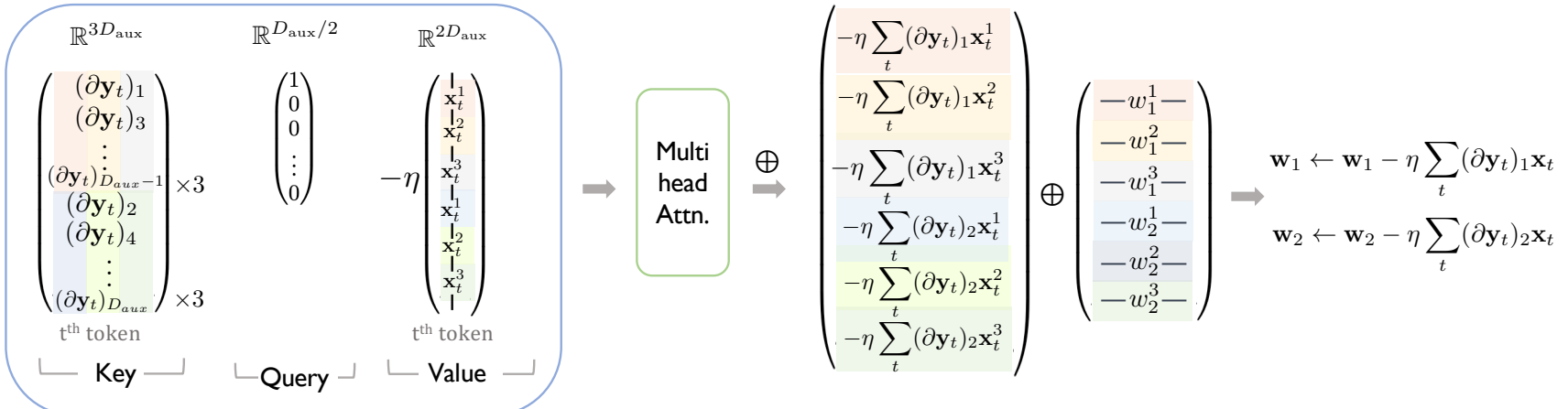
$\mathbb{R}^{3D_{\text{aux}}}$
$\mathbb{R}^{D_{\text{aux}}/2}$
$\mathbb{R}^{2D_{\text{aux}}}$
$(\partial \mathbf{y}_t)_1$
$(\partial \mathbf{y}_t)_3$
$(\partial \mathbf{y}_t)_{D_{aux}-1}$
$\times 3$
$(\partial \mathbf{y}_t)_2$
$(\partial \mathbf{y}_t)_4$
$(\partial \mathbf{y}_t)_{D_{aux}}$
$\times 3$
1
0
0
0
$-\eta$
$\mathbf{x}_t^1$
$\mathbf{x}_t^2$
$\mathbf{x}_t^3$
$\mathbf{x}_t^1$
$\mathbf{x}_t^2$
$\mathbf{x}_t^3$
t[th] token
t[th] token
Key
Query
Value
Multi head Attn.
$\oplus$
$-\eta \sum_t (\partial \mathbf{y}_t)_1 \mathbf{x}_t^1$
$-\eta \sum_t (\partial \mathbf{y}_t)_1 \mathbf{x}_t^2$
$-\eta \sum_t (\partial \mathbf{y}_t)_1 \mathbf{x}_t^3$
$-\eta \sum_t (\partial \mathbf{y}_t)_2 \mathbf{x}_t^1$
$-\eta \sum_t (\partial \mathbf{y}_t)_2 \mathbf{x}_t^2$
$-\eta \sum_t (\partial \mathbf{y}_t)_2 \mathbf{x}_t^3$
$\oplus$
$-w_1^1-$
$-w_1^2-$
$-w_1^3-$
$-w_2^1-$
$-w_2^2-$
$-w_2^3-$
$\mathbf{w}_1 \leftarrow \mathbf{w}_1 - \eta \sum_t (\partial \mathbf{y}_t)_1 \mathbf{x}_t$
$\mathbf{w}_2 \leftarrow \mathbf{w}_2 - \eta \sum_t (\partial \mathbf{y}_t)_2 \mathbf{x}_t$